Project: Worked examples
Title: Shaft under combined bending and torsion
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Bending and torsional stresses in a solid round shaft with stress-concentration factors, the maximum shear (Tresca) and von Mises equivalent stresses, and the factors of safety against yield, against a required factor.

$\sigma = K_t \cdot 32M/(\pi d^3)$; $\tau = K_{ts} \cdot 16T/(\pi d^3)$; $\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)}$; $\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$; $n_T = (S_y/2)/\tau_{max}$; $n_{VM} = S_y/\sigma'$. The lower factor governs.

Ref: Shigley's Mechanical Engineering Design §5-4, §5-5 — Maximum-shear-stress and distortion-energy theories
Ref: Shigley's Mechanical Engineering Design §7-4 — Shaft design for stress

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Shaft diameter	25	mm	drawing: DRG-SH-402 rev A	1 … 2000 mm
M	Bending moment (magnitude)	120	N*m	spec: LOAD-SCHEDULE-01	0 … 10000000 N*m
T	Torque (magnitude)	60	N*m	spec: LOAD-SCHEDULE-01	0 … 10000000 N*m
S_y	Yield strength	250	MPa	drawing: DRG-SH-402 rev A	50 … 3000 MPa
K_t	Bending stress-concentration factor (1 for none)	1.6		spec: SPEC-MECH-03 design criteria	1 … 5
K_ts	Torsional stress-concentration factor (1 for none)	1.3		spec: SPEC-MECH-03 design criteria	1 … 5
n_req	Required factor of safety	1.5		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. Solid round section, linear elastic; static loading. (Fatigue is mech.fatigue-miner's job; hollow shafts are not covered.)

4. Working

```
[8] sigma = K_t * 32 * M / (pi * d^3)                    Shigley §7-4
= 1.6 × 32 × 120 N·m / (π × (25 mm)^3)
= 125.16 MPa
```

Bending stress

```
[9] tau = K_ts * 16 * T / (pi * d^3)                     Shigley §7-4
= 1.3 × 16 × 60 N·m / (π × (25 mm)^3)
= 25.424 MPa
```

Torsional shear stress

[10] tau_max = sqrt((sigma / 2)^2 + tau^2) Shigley §5-4
= sqrt((125.16 MPa / 2)^2 + (25.424 MPa)^2)
= 67.549 MPa
Maximum shear stress (Tresca)
[11] sigma_vm = sqrt(sigma^2 + 3 * tau^2) Shigley §5-5
= sqrt((125.16 MPa)^2 + 3 × (25.424 MPa)^2)
= 132.68 MPa
von Mises stress
[12] n_T = (S_y / 2) / tau_max
= 250 MPa / 2 / 67.549 MPa
= 1.8505
Factor of safety, maximum shear stress
[13] n_VM = S_y / sigma_vm
= 250 MPa / 132.68 MPa
= 1.8842
Factor of safety, distortion energy
[14] n_min = min(n_T, n_VM)
= min(1.8505, 1.8842)
= 1.8505
Governing factor of safety
[15] u = n_req / n_min
= 1.5 / 1.8505
= 0.81059
Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
sigma	125.16 MPa	Bending stress
tau	25.424 MPa	Torsional shear
tau_max	67.549 MPa	Maximum shear stress
sigma_vm	132.68 MPa	von Mises stress
n_T	1.8505	Factor of safety (Tresca)
n_VM	1.8842	Factor of safety (von Mises)
n_min	1.8505	Governing factor of safety

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	u = 0.81059	u ≤ 1	0.811	Shigley §7-4

7. Applicability

No warnings.

8. Verification

Template mech.shaft-stress@1.0.0 Content hash 6564629703e39ada Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0